

External Data Ablation Report

加入 Weather、Building Footprints、Building Permits 與 All External Features 的成果比較

1. 目的

比較三種外部資料與三者合併後，是否能提升 Chicago Energy Benchmarking 的 Site EUI regression model。

2. 實驗設定

Train = 2018-2021 ; Validation = 2022，用於 random search hyperparameter tuning ; Test = 2023，只做最後評估。模型包含 Random Forest Regressor 與 XGBoost Regressor，評估指標為 RMSE、MAE、 R^2 。

3. Feature Engineering 摘要

Base features 包含歷史 EUI、community/property type historical mean、building age、log floor area、community-year count 等。Weather 加入 HDD/CDD、溫度、降水與極端溫度天數；Footprints 加入建築 footprint 面積、周長、compactness、樓層與 footprint age；Permits 加入前三年 permit count、cost、HVAC/electrical/renovation 類型與 years since last permit。

3.1 主要 Feature 公式

Base feature engineering

Feature	Formula / Definition
years_since_2018	data_year - 2018
log_floor_area	$\log(1 + \text{gross_floor_area_buildings_sq_ft})$
building_age	data_year - year_built; invalid if < 0 or > 250
regulation_stage	2018-2019 pre-covid; 2020-2021 covid; 2022-2023 post-covid
prev_year_site_eui	same building's Site EUI at year t-1
prior_building_mean_eui	mean Site EUI of same building for years < t
prior_community_mean_eui	mean Site EUI of same community for years < t
prior_property_type_mean_eui	mean Site EUI of same building type for years < t
prior_comm_property_mean_eui	mean Site EUI of same community + building type for years < t
prior_community_high_eui_ratio	proportion of records with Site EUI >= 100 in same community for years < t
community_year_record_count	number of records in the same community and year
community_year_building_count	number of unique buildings in the same community and year
community_year_avg_floor_area	average gross floor area in the same community and year
property_year_record_count	number of records in the same property type and year

Weather feature engineering

Feature	Formula / Definition
weather_temp_mean_c	annual mean of daily mean temperature
weather_temp_max_mean_c	annual mean of daily maximum temperature
weather_temp_min_mean_c	annual mean of daily minimum temperature
weather_precip_sum_mm	annual sum of daily precipitation
weather_hdd65	$\text{sum}(\max(18.33 - \text{daily_mean_temp_c}, 0))$
weather_cdd65	$\text{sum}(\max(\text{daily_mean_temp_c} - 18.33, 0))$
weather_hot_days_30c	count of days with daily max temperature >= 30 C
weather_cold_days_minus10c	count of days with daily min temperature <= -10 C

Building footprint feature engineering

Feature	Formula / Definition
footprint_matched	1 if address matched to building footprint, otherwise 0
footprint_shape_area	footprint polygon area
footprint_shape_len	footprint polygon perimeter
footprint_compactness	$4 * \pi * \text{shape_area} / \text{shape_len}^2$
footprint_stories	number of stories from footprint data
footprint_year_built	year built from footprint data
footprint_bldg_sqft	building square footage from footprint data
gross_to_footprint_area_ratio	$\text{gross_floor_area_buildings_sq_ft} / \text{footprint_shape_area}$
footprint_age	$\text{data_year} - \text{footprint_year_built}$

Building permit feature engineering

Feature	Formula / Definition
permit_matched	1 if the address had permit history before year t, otherwise 0
permit_count_3yr	permit count from t-3 to t-1
permit_count_all_prior	all permit count before t
permit_reported_cost_sum_3yr	sum of reported cost from t-3 to t-1
permit_total_fee_sum_3yr	sum of permit fees from t-3 to t-1
permit_mechanical_count_3yr	count of HVAC/mechanical/heating/cooling permits from t-3 to t-1
permit_electrical_count_3yr	count of electrical/wiring/lighting permits from t-3 to t-1
permit_renovation_count_3yr	count of renovation/alteration/rehab/repair permits from t-3 to t-1
permit_major_cost_count_3yr	count of permits with reported_cost $\geq$ 100,000 from t-3 to t-1
years_since_last_permit	$\text{data_year} - \text{last_permit_year}$
has_recent_permit	1 if permit_count_3yr > 0, otherwise 0

4. Join Coverage

feature_set	matched_rows	matched_row_rate	matched_unique_addresses	unique_addresses	recent_permit_rows	recent_permit_row_rate	weather_matched_rows	weather_matched_row_rate	footprint_matched_rows	footprint_matched_row_rate	permit_matched_rows	permit_matched_row_rate
base	15177	1.0000										
weather	15177	1.0000										
footprints	11481	0.7565	3085.0000	4248.0000								
permits	10706	0.7054			9613.0000	0.6334						
all_external	15177	1.0000			9613.0000	0.6334	15177.0000	1.0000	11481.0000	0.7565	10706.0000	0.7054

5. Hyperparameter Tuning

每個 feature set 都重新跑 random search。Random Forest 跑 20 組，XGBoost 跑 25 組，並用 validation RMSE 選最佳參數。

```
Random Forest:
n_estimators = [200, 300, 500]
max_depth = [8, 12, 16, 24, None]
min_samples_leaf = [1, 3, 5, 10]
max_features = ['sqrt', 0.5, 0.8]

XGBoost:
n_estimators = [200, 400, 600]
max_depth = [2, 3, 4, 5, 6]
learning_rate = [0.03, 0.05, 0.08, 0.1]
subsample = [0.7, 0.85, 1.0]
colsample_bytree = [0.7, 0.85, 1.0]
reg_lambda = [1, 5, 10]
min_child_weight = [1, 3, 5]
```

Feature Set	Model	Iter	Val RMSE	Val MAE	Val R ²
base	XGBoost	7	21.1808	12.3151	0.8067
weather	Random Forest	18	21.4582	11.9469	0.8016
footprints	XGBoost	11	20.9459	12.0954	0.8110
permits	XGBoost	7	20.7603	12.2395	0.8143
all_external	Random Forest	18	21.4591	11.9103	0.8016

Validation-selected best hyperparameters

```
{
  "base": {
    "model": "XGBoost",
    "params": {
      "colsample_bytree": 0.85,
      "learning_rate": 0.1,
      "max_depth": 6,
      "min_child_weight": 1,
      "n_estimators": 400,
      "reg_lambda": 5,
      "subsample": 0.7
    }
  },
  "weather": {
    "model": "Random Forest",
    "params": {
      "max_depth": 24,
      "max_features": 0.5,
      "min_samples_leaf": 3,
      "n_estimators": 500
    }
  },
  "footprints": {
    "model": "XGBoost",
    "params": {
      "colsample_bytree": 0.7,
      "learning_rate": 0.05,
      "max_depth": 6,
      "min_child_weight": 3,
      "n_estimators": 400,
      "reg_lambda": 5,
      "subsample": 0.85
    }
  }
}
```

```

    }
  },
  "permits": {
    "model": "XGBoost",
    "params": {
      "colsample_bytree": 0.85,
      "learning_rate": 0.1,
      "max_depth": 6,
      "min_child_weight": 1,
      "n_estimators": 400,
      "reg_lambda": 5,
      "subsample": 0.7
    }
  },
  "all_external": {
    "model": "Random Forest",
    "params": {
      "max_depth": 24,
      "max_features": 0.5,
      "min_samples_leaf": 3,
      "n_estimators": 500
    }
  }
}

```

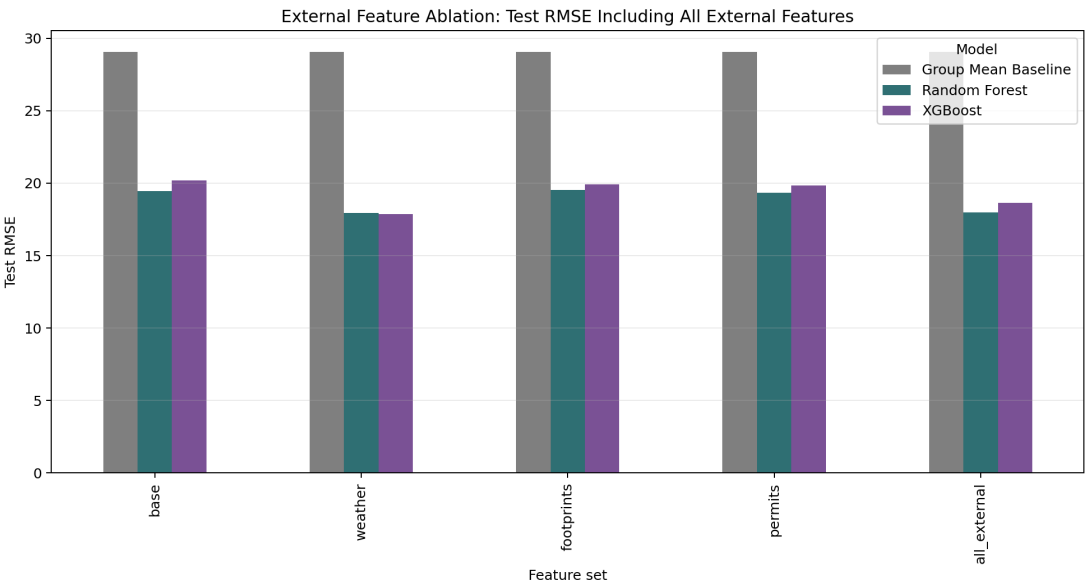
6. Test Results

Feature Set	Model	RMSE	MAE	R ²
all_external	Random Forest	17.9975	11.0789	0.8208
all_external	XGBoost	18.6474	11.6705	0.8076
all_external	Group Mean Baseline	29.0747	20.4876	0.5323
base	Random Forest	19.4437	13.1277	0.7908
base	XGBoost	20.1889	13.8336	0.7745
base	Group Mean Baseline	29.0747	20.4876	0.5323
footprints	Random Forest	19.5151	13.0831	0.7893
footprints	XGBoost	19.9053	13.2463	0.7808
footprints	Group Mean Baseline	29.0747	20.4876	0.5323
permits	Random Forest	19.3409	12.9813	0.7930
permits	XGBoost	19.8375	13.6813	0.7823
permits	Group Mean Baseline	29.0747	20.4876	0.5323
weather	XGBoost	17.8748	11.0647	0.8232
weather	Random Forest	17.9485	11.0946	0.8218
weather	Group Mean Baseline	29.0747	20.4876	0.5323

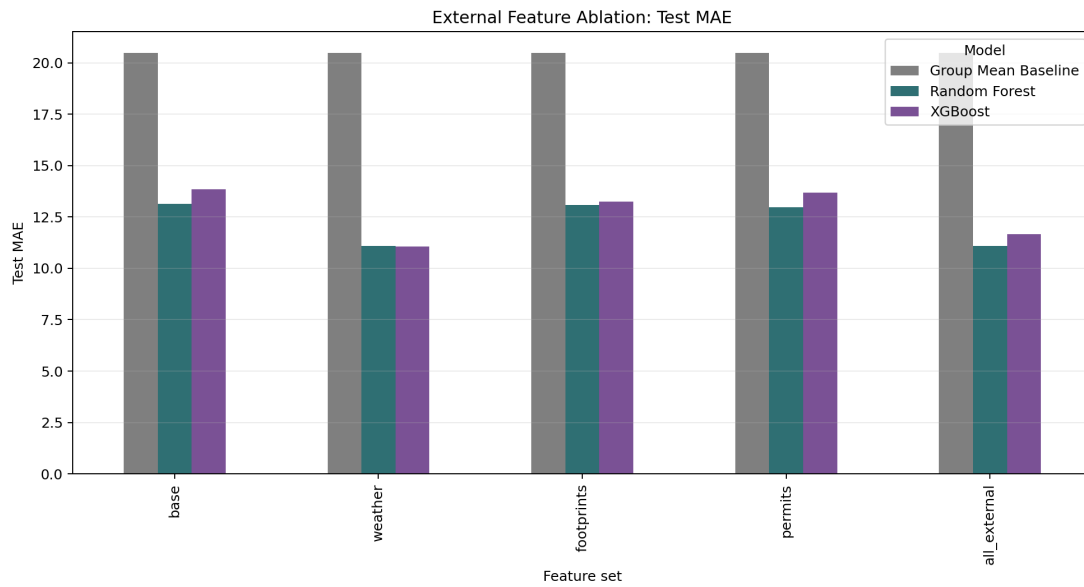
7. Best Model By Feature Set

Feature Set	Best Model	RMSE	MAE	R ²	ΔRMSE vs Base	ΔRMSE vs Weather
base	Random Forest	19.4437	13.1277	0.7908	0.0000	1.5689
weather	XGBoost	17.8748	11.0647	0.8232	-1.5689	0.0000
footprints	Random Forest	19.5151	13.0831	0.7893	0.0714	1.6403
permits	Random Forest	19.3409	12.9813	0.7930	-0.1028	1.4661
all_external	Random Forest	17.9975	11.0789	0.8208	-1.4462	0.1227

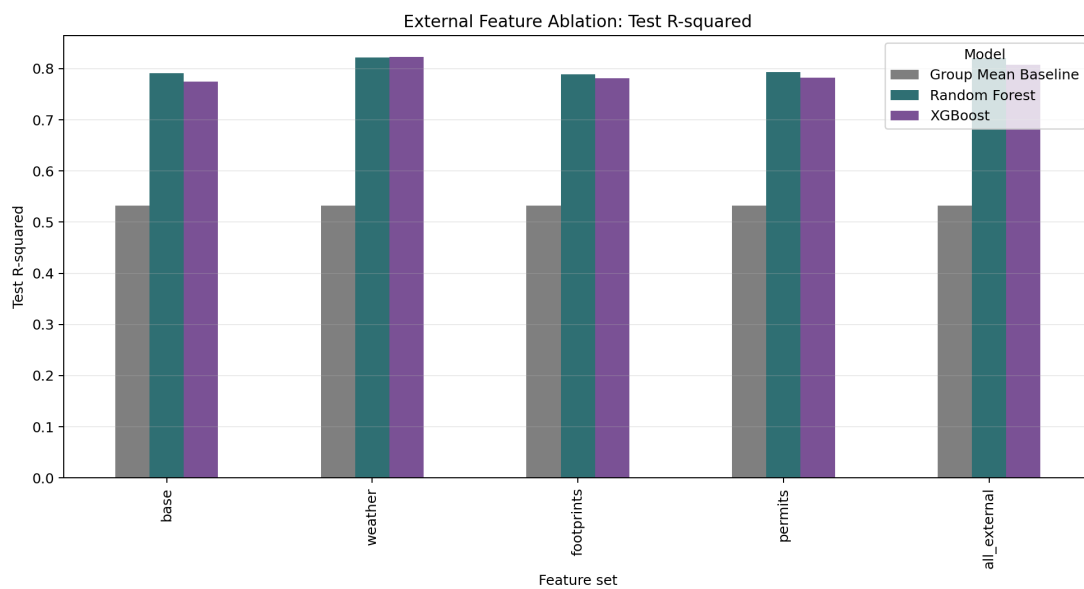
RMSE comparison



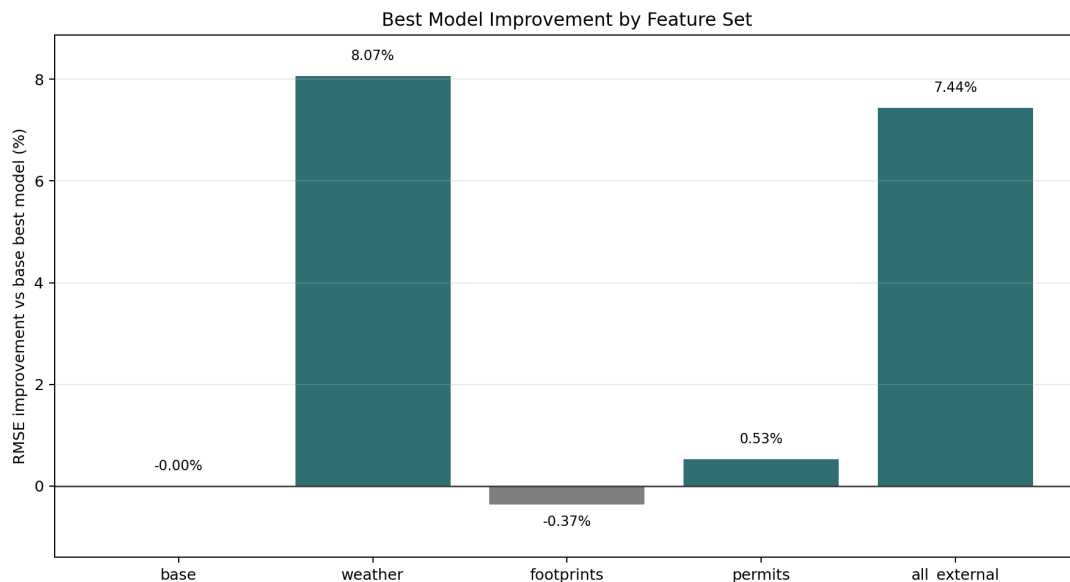
MAE comparison



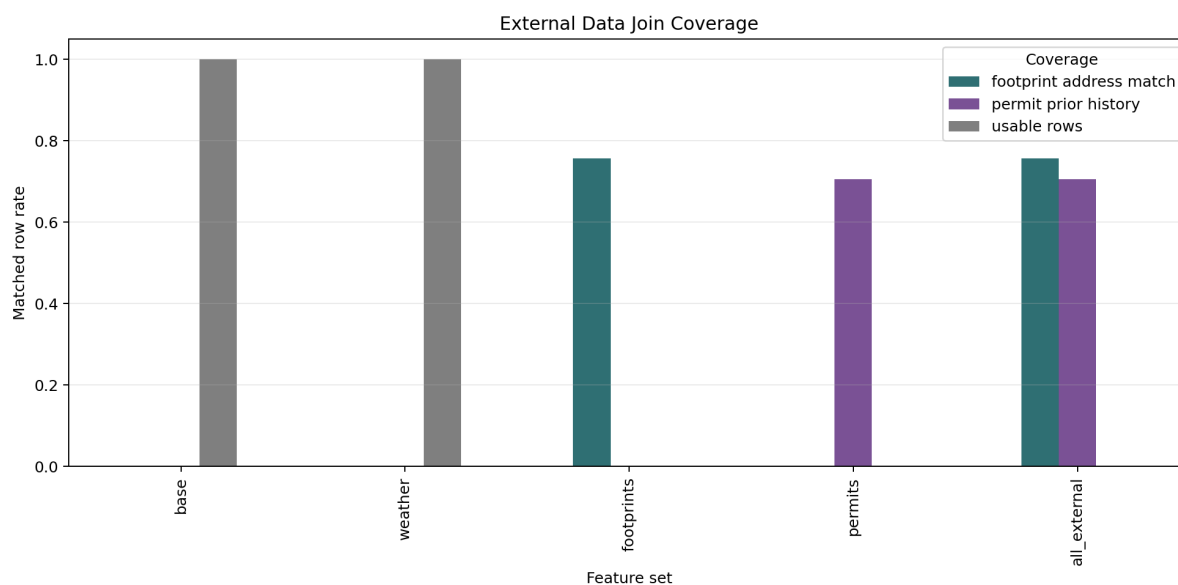
R-squared comparison



Best model RMSE improvement



External data join coverage



8. 成果解讀

結果顯示 weather-only 是最佳設定。All external 比 base 好，但沒有超過 weather-only，代表 footprints 與 permits 在目前地址匹配與特徵設計下，沒有在 weather 已存在時提供穩定額外訊號。

9. 結論

最終建議模型為 Base features + Weather features + XGBoost Regressor。這個模型在 test set 上取得最低 RMSE 與最高 R^2 ，且比 all external 更簡潔。